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INVENTORY MODELLING

Inventory modeling deals with determining the level of a commodity that a business must maintain to ensure smooth operation. Inventories are vital to the successful functioning of manufacturing and retailing organizations. The inventories may consist of raw materials, work-in-progress, spare parts / consumables, and finished goods.

Inventories may be held for a variety of purposes, but in general, there are five types of inventories that an organization can use for serving these purposes. These are

a) Movement Inventories

— Also called transit or pipeline inventories. Their existence owes to the fact that transportation time is involved in transferring substantial amounts of resources.

b) Buffer Inventories

— Are held to protect against the uncertainties of demand and supply. An organization generally knows the average demand for various items that it needs. However, the actual demand may not exactly match the average and could well exceed it.

Note:

Lead time:

Time elapsing between placing an order and having the goods in stock ready for use.

Safety stocks are held for protecting against the fluctuations in demand and lead time. The idea of keeping buffer is to render a higher level of customer service and [illegible] reduce the [illegible] of stockouts and back-orders.

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(c)

Anticipating Inventories

—They are held for the reason that a future demand the product is anticipated. The idea underlying is to Smoothing the production process producing for a longer duration on a continuous basis rather than operating with excessive overtime in the planned and then let the system be idle or close down for the reason of inadequate demand for another period.

(d) Decoupling Inventories

—The idea of it is to decouple or disengage different parts of the production system. Clearly the decoupling inventories act as shock absorbers and have cushion effect in the face of varying work-rates, and machine breakdowns and failures, and so on.

(e) Cycle Inventories

— They are held for the reason that purchases are made in lots rather than the exact amounts which they be needed at a point in time.
As the cycle inventories are related to the purchase in lots, they are also called lot-size inventories and the larger the lot size the greater would be the level of cycle inventories.

General Inventory Model / Decision

In an inventory control situations, there are the basic questions to be answered & they are:

order?

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The bases for considering these decisions is the minimization of the following inventory cost function:

$$\begin{pmatrix}\text{Total}\\ \text{Inventory}\\ \text{cost}\end{pmatrix} = \begin{pmatrix}\text{Purchasing}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Setup}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Holding}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Shortage}\\ \text{cost}\end{pmatrix}$$

Inventory management systems

There are two inventory management systems. They are:

Fixed order quantity system (or reorder point system)

In which a level called the re-order level is determined. As soon as the stock level reaches this point, an order for a predetermined number of units is placed.

Periodic review system

where the inventories are replenished at fixed intervals of time. Whereas the size of order is fixed in the previous system and the time is not, in this system the time after which the supplies are ordered is fixed but not the quantity to be ordered.

Demand and inventory models

The demand pattern in an inventory model may assume one of four types:

Deterministic and constant (static) with time

Deterministic and variable (dynamic) with time

probabilistic and stationary over time

probabilistic and non-stationary over time

To determine if inventory model is a fair approximation of demand

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use it to determine the nature of demand using the following guideline

1. If the average monthly demand is "approximately" constant for all months and CV is reasonably small (< 20%), then the demand may be considered deterministic and can with the value equal to the average of all monthly demands

2. If the average monthly demand varies appreciably among different months but CV remains reasonably small, the the demand is considered deterministic but varies

ANOVA test

3. If, in case 2, CV is high (> 20%) and approximately con then the demand is probabilistic and stationary.

4. If the ~~[illegible]~~ climatic is is probabilistic and non-stationary.

Example

Analyse the nature of the following demand for product X

Weekly Demand

months	week 1	2	3	4
Jan	43	44	45	68
Feb	69	90	41	79
Mar	50	76	34	81
April	64	80	35	22
May	27	86	41	80
June	34	74	40	20

Sigma

mean

S.d DEV

	wk1	wk2	wk3	wk4	mean	std deviation	COV
Jan	43	99	45	68	50.00	10.42	20.83
Feb	69	90	41	79	69.75	18.18	26.07
Mar	50	76	34	81	60.25	19.19	31.85
Apr	67	80	39	22	52.00	22.79	43.83
May	27	86	41	80	58.50	25.08	42.88
Jun	34	74	40	20	42.00	19.85	47.26
Solution							
mean	48.33	75.00	40.00	58.33			
std deviation	15.64	14.91	3.27	26.75			
cov	32.36	19.88	8.16	45.86			

SUMMARY	Count	Sum	Average	Variance
Jan	4	200	50	144.6667
Feb	4	279	69.75	440.9167
Mar	4	241	60.25	490.9167
Apr	4	208	52	692.6667
May	4	234	58.5	839
Jun	4	168	42	525.3333
wk1	6	290	48.33333	293.4667
wk2	6	450	75	266.8
wk3	6	240	40	12.8
wk4	6	350	58.33333	858.6667

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Months	17.333	5	367.4667	1.035831	0.432306	2.901295
Weeks	19.167	3	1359.722	3.832843	0.032085	3.287382
Error	1.333	15	354.7556			
Total	37.83	23				

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Inventory Costs

In determining an optimal inventory policy, all or some of these cost components are included in the objective function

Purchase Cost

This refers to the nominal cost of inventory. It is the purchase price for the items that are sought from outside sources and the production cost of items are produced within the organization

Ordering Cost / Setup Cost

This is incurred whenever the inventory is replenished. It includes costs associated with the processing and clearing of the purchase order, transportation, inspection for quality, expediting overtime etc. It is also known as procurement cost.

Carrying Cost

Also known as holding or storage cost, carrying cost represents the cost that is associated with the processing and storing an item in inventory. Its elements include the opportunity cost of capital at is invested in the stock; the costs directly associated with storing goods; the obsolescence cost; deterioration costs and cost incurred in preventing deteriorations, and fire and general insurance etc.

Stockout Cost

The cost associated with not serving the customers. Stockouts imply shortages. This could be internal or external.

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Fixed-order Quantity System

The classical EOQ Model (Wilson formulation)

EOQ ——— Economic order quantity model

Under the EOQ model, a cost model is first developed, and then it is manipulated to form an inventory model.

Assumptions

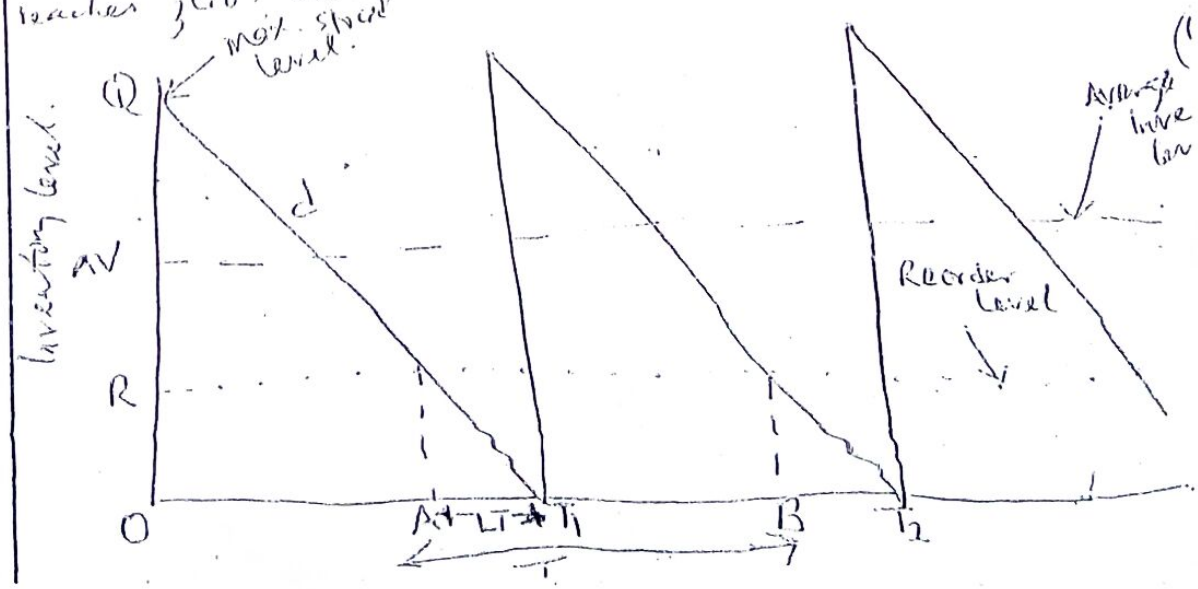
i. The demand for the item is certain, continuous and consta... with time

ii. The lead time, that is, the time between placing an order and its delivery is known and fixed. Thus when the lead time is zero, the delivery is instantaneous.

iii. Within the range of the quantities to be ordered, the per unit holding cost and the ordering cost are constant and independent of the quantity ordered.

iv. The per unit purchase price of the item is constant, no discount is available on purchase of large lots.

v. The inventory is replenished immediately as the stock level reaches zero. Consequently, there are no stock overages or shortages



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The cost Model is

$$T(Q) = O(Q) + H(Q)$$

where Q — the ordering quantity

$T(Q)$ — total (variable) annual inventory cost

$O(Q)$ — total annual ordering cost

$H(Q)$ — total annual holding cost.

Example

Ada Ltd produces 2000 TV sets in a year for which it needs an equal number of tubes of a certain type. Each tube costs ₦10 (ten) and the cost of storing a tube in stock for a year is ₦2.40. Besides, the cost of placing an order is ₦150, which is not related to its size.

Now, if an order for 2,000 tubes is placed only one order per annum is required. When 1000 units are ordered for, 2 orders in a year are needed, while if 500 units are ordered to be supplied, then a total of 4 orders per annum are required. Naturally, as the number of orders placed increases the ordering cost goes up. More orders, however, would also imply smaller order quantity and, therefore, decreasing holding costs. Thus we have here a trade-off between the ordering and the holding cost.

Find the order size that minimizes the cost function $T(Q)$.

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Sol:

(a) Total annual ordering cost

This is given by the number of times an order is placed, multiplied by the ordering cost per order, A.

i.e. $O(Q) = N \times A$

Where $N = \frac{D}{Q} = \frac{\text{annual demand}}{\text{order quantity}}$

$$O(Q) = \frac{D}{Q} \cdot A$$

For the example when

$N = 1$, $Q = 2000$ and $O(Q) = 1 \times 150 = 150$

$N = 2$, $Q = 1000$ and $O(Q) = 2 \times 150 = 300$

$N = 3$, $Q = 500$ and $O(Q) = 3 \times 150 = 450$

$N = 5$, $Q = 400$ and $O(Q) = 5 \times 150 = 750$

(b) Total annual holding cost

Obtained by multiplying the unit holding cost, h, by the average number of units held in the inventory, $Q/2$

$$H(Q) = \frac{Q}{2} \cdot h$$

For our example, when

$Q = 2000$, $H(Q) = \frac{2000}{2} \times 2.40 = 2400$

$Q = 1000$, $H(Q) = \frac{1000}{2} \times 2.40 = 1200$

$Q = 500$, $H(Q) = \frac{500}{2} \times 2.40 = 600$

$Q = 400$, $H(Q) = \frac{400}{2} \times 2.40 = 480$

Note

If i is the proportion (or percentage) which is called the holding rate, and c is the cost of the item, then

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Total annual inventory cost (Variable) obtained by adding up both components of EQ and PQ.

Calculation of Total cost

Order quantity	Annual ordering cost	Annual Holding cost	Total Annual cost
100	3000	120	3120
200	1500	240	1740
400	750	480	1230
500	600	600	1200
1000	300	1200	1500
2000	150	2400	2550

Determine of the re-order level

Suppose the following additional data are given

No of working days in the year = 250

Lead time = 15 working days

With this information, the daily demand = 2000/250 = 8

Demand during lead time = 15 × 8 = 120

∴ Re-order level = 120

Annual total variable inventory cost

The minimum annual inventory cost can be determined by substituting Q^* for Q in the cost equation, Thus

$$T(Q^*) = \frac{D}{Q^*}A + \frac{Q^*}{2}h$$

$$= \frac{DA}{\sqrt{\frac{2AD}{h}}} + \frac{\left(\sqrt{\frac{2AD}{h}}\right)h}{2}$$

$$\Rightarrow T(Q^*) = \sqrt{2AD}$$

∴ for the example

$$T(Q^*) = \sqrt{2 \times 150 \times 2000 \times 2.40}$$

$$= \sqrt{1440000} = 1200$$

Example

Using the following data, obtain the EOQ and the total variable cost associated with policy of ordering quantities of the size:

Annual Demand = 20,000

ordering cost = 150 per order

Inventory carrying cost = 24% of average Inventory value

Soln

$CD = ₦20,000$, C — unit price

$A = 150$/order D — demand

$i = 24\% = 0.24$

$$EOQ = Q = \sqrt{\frac{2 \times 150 \times 20,000}{0.24}}$$

$$= 5000$$

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$$EOQ = \sqrt{\frac{2ACD}{i}}$$

Total cost

$$T(Q^*) = \sqrt{2ACDi}$$

C.D — the annual demand

A — the acquisition cost

i — the holding rate.